

Week 4 Report: Shopify Stock Performance Analysis

1. Introduction

- This project focuses on building an interactive Power BI dashboard for analyzing Shopify stock data. The dashboard integrates timeline slicers, dynamic metric switching, and custom tooltips to provide a professional and user-friendly analytical experience.

Step 1: Open Week 3 File

- Start by opening your Week 3 Power BI file.
- This ensures continuity and keeps all dashboards in one .pbix file.

Step 2: Add a New Page

- Click the “+” icon at the bottom → create a new report page.

Step 3: Create New Measures (in the existing measures table)

Calculate SMA(Simple Moving Average) for 20 days and 50 days using DAX

- `Volume 20 SMA = VAR CurrentDate = MAX('Date Table'[Date])
VAR PeriodDays = 20
VAR DateInPeriod = DATESINPERIOD('Date Table'[Date], CurrentDate, -PeriodDays, DAY)
RETURN CALCULATE(AVERAGE('SHOP_2015-05-21_2025-03-16'[Volume]), DateInPeriod)`
- `Volume 50 SMA = VAR CurrentDate = MAX('Date Table'[Date])
VAR PeriodDays = 50
VAR DateInPeriod = DATESINPERIOD('Date Table'[Date], CurrentDate, -PeriodDays, DAY)
RETURN CALCULATE(AVERAGE('SHOP_2015-05-21_2025-03-16'[Volume]), DateInPeriod)`

Moving Average Crossover Signal (Golden / Death Cross Indicator)

- **MA Crossover Signal = IF(ISBLANK([Volume 20 SMA]) || ISBLANK([Volume 50 SMA]), BLANK(), IF([Volume 20 SMA] > [Volume 50 SMA], "Bullish (20 > 50)", "Bearish (20 < 50)"))**

Step 4: Add Line Chart with SMA & Close Price

- Insert a Line Chart on the new Week 4 page.
- Drag Dim_Date[Date] into the X-axis.
- Add the following measures into Values:
 - Volume 20 SMA
 - Volume 30 SMA
 - Close Price
- Go to Format → X-axis → Continuous to ensure smooth timeline.
- Enable Zoom Slider for both X-axis and Y-axis to allow interactive zooming.



👉 This chart shows how short-term and medium-term averages compare with the actual closing price.

Step 5: Add Line Chart with SMA & Volume

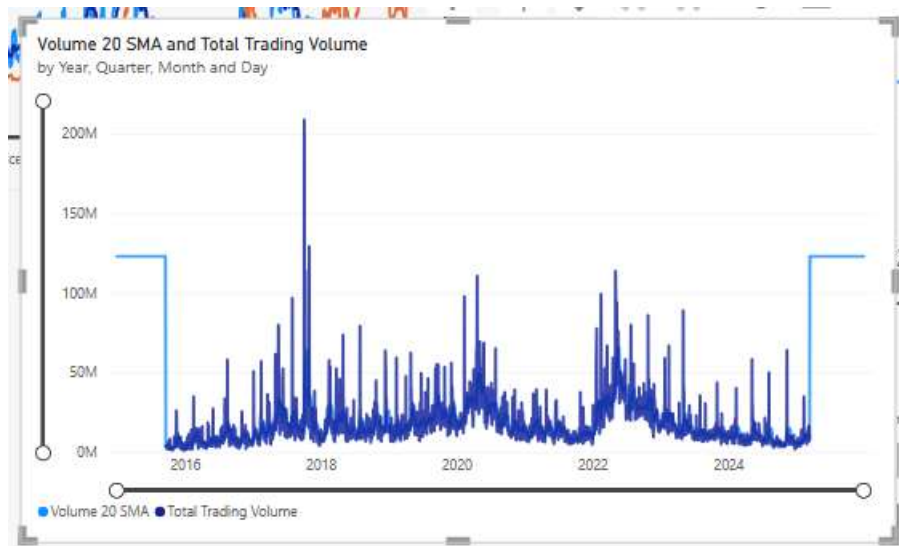
- Insert a Line Chart on the Week 4 page.
- Drag Dim_Date[Date] into the X-axis.

- Add the following measures into Values:

Volume 20 SMA

Total Trading Volume

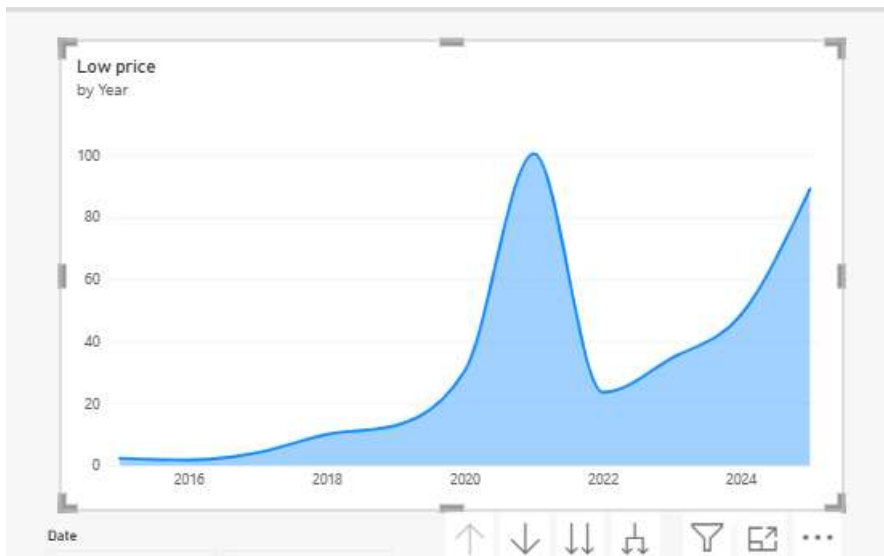
- Go to Format → X-axis → Continuous to ensure smooth timeline.
- Enable Zoom Slider for both X-axis and Y-axis to allow interactive zooming.



☞ This chart shows how actual trading volume compares against the 20-day moving average.

Step 6: Add Area Chart with Low Price & Parameter Field

- Insert a new Area Chart in the Week 4 page.
- Drag Dim_Date[Date] into the X-axis.
- Add the following measure into Values:
Low Price
- Go to Modeling → New Parameter → Fields.
- Selected measures: Open Price, High Price, Low Price, Close Price.
- Power BI automatically created:
A parameter slicer (to switch between Open, High, Low, Close).



👉 Now the Area Chart can dynamically update based on whichever metric (Open, High, Low, Close) is selected in the slicer.

Step 7: Add Date Slicer

- Go to the Visualizations pane → select the Slicer icon (funnel-shaped rectangle).
- Drag Dim_Date[Date] into the slicer field.
- In the slicer settings, change it to Between mode.
- This shows two boxes → Start Date and End Date.



👉 With this, users can filter the dashboard either by **exact date ranges** or by **relative periods**.

Step 8: Create Custom Tooltip Page

- Add a new report page → rename it Stock_Tooltip.
- Go to Format Page → Page Information → Allow as Tooltip = On.
- In Canvas Settings → Type = Tooltip.

👉 This makes the page act as a tooltip instead of a normal report page.

- Insert Card visuals for:

Open Price

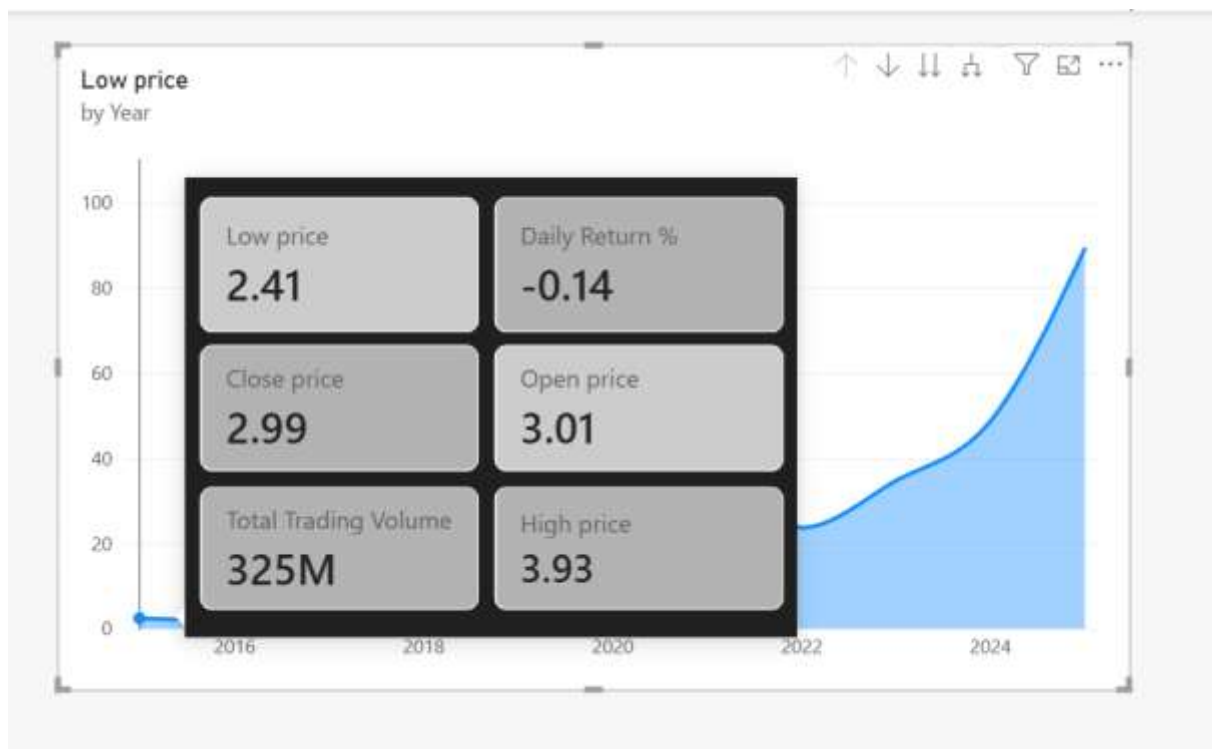
High Price

Low Price

Close Price

Step 9: Connect Tooltip Page to Area Chart

- Click the Area Chart you created earlier (with Low Price + parameter field).
- Go to the Format pane → General → Tooltip.
- In the Tooltip settings, select Page → Page 3 (or whichever page you prepared for detailed metrics).
- Now, when you point to / move your cursor over any data point in the Area Chart, the tooltip from Page 3 will appear.
- It will show: Open Price, High Price, Low Price, Close Price, Total Trading Volume, Daily Change %.



Final output:

